

Preparation of Standard Curve of Proteins using Lowry's Technique and Estimation of Protein from an Unknown Sample

Apparatus and Materials Required

- Spectrophotometer
- Cuvettes
- Pipettes
- Test tubes
- Vortex mixer
- Distilled water
- Bovine Serum Albumin (BSA) (standard protein)
- Sodium carbonate (Na_2CO_3) solution (2% w/v)
- Copper sulfate (CuSO_4) solution (0.5% w/v)
- Folin-Ciocalteu reagent
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Procedure

1. Prepare BSA standard dilutions:

- Prepare a stock solution of BSA at a known concentration (1mg/mL).
- Using distilled water, prepare a series of dilutions of the BSA stock solution to obtain the following range of standard concentrations: 0 μg (blank), 10 μg , 20 μg , 30 μg , 40 μg , and 50 μg .

2. Reaction:

- Add 500 μL of sodium carbonate solution (2% w/v) to each test tube and mix by vortexing. Incubate for 10 minutes at room temperature.
- Add 100 μL of copper sulfate solution (0.5% w/v) to each test tube and mix by vortexing. Incubate for another 10 minutes at room temperature.
- Finally, add 50 μL of Folin-Ciocalteu reagent (diluted) to each test tube and mix thoroughly by vortexing.
- Add water to make the total volume of the sample 1 mL. Incubate for 30 minutes at room temperature in the dark.

3. Measurement:

- After incubation, adjust the spectrophotometer to 650 nm wavelength.
- Using a cuvette filled with distilled water as the blank, measure the absorbance of each standard solution at 650 nm.

4. Plotting the Standard Curve

Using a graph paper, plot a standard curve with BSA (μg) on the x-axis and absorbance (650 nm) on the y-axis. This standard curve will be used to determine the protein concentration of unknown samples using their absorbance values in subsequent experiments.

5. Estimation of protein from an unknown sample

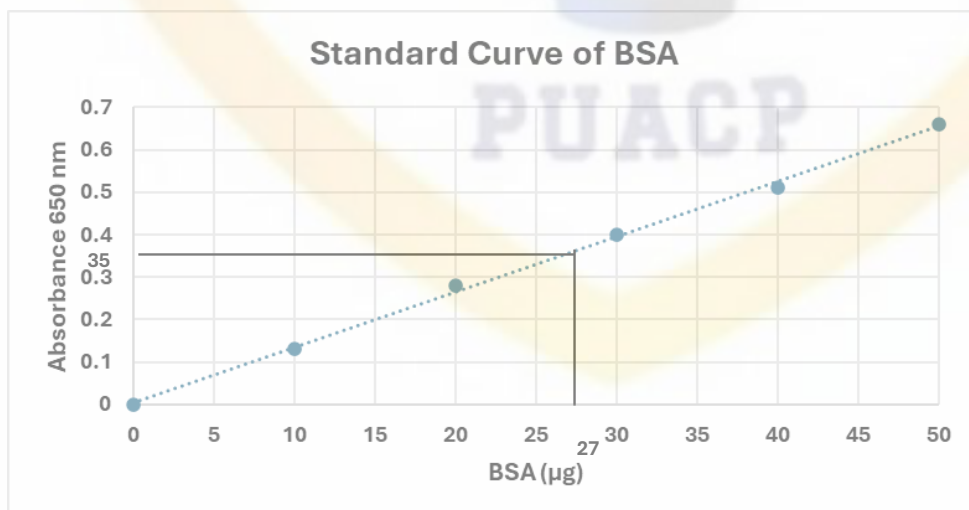
- Add 50 μL of unknown sample in a test tube and add the above mentioned reagents in the tube.
- Using a cuvette filled with distilled water as the blank, measure the absorbance of unknown solution at 650 nm.

Results

The absorbance of unknown solution was 0.35 which corresponds to 27 μg of BSA according to the standard curve of BSA.

Sr. No.	Stock Solution (μL)	H ₂ O (μL)	Na ₂ CO ₃ (μL)	CuSO ₄ (μL)	Folin-Ciocalteu reagent (μL)	BSA (μg)	Absorbance (650 nm)
1	0	350	500	100	50	0	0
2	10	340	500	100	50	10	0.13
3	20	330	500	100	50	20	0.28
4	30	320	500	100	50	30	0.4
5	40	310	500	100	50	40	0.51
6	50	300	500	100	50	50	0.66

Sr. No.	Unknown Solution (μL)	H ₂ O (μL)	Na ₂ CO ₃ (μL)	CuSO ₄ (μL)	Folin-Ciocalteu reagent (μL)	BSA (μg)	Absorbance (650 nm)
1	50	300	500	100	50	27	0.35



Preparation of Standard Curve and Estimation of DNA by Colorimetric Analysis using Diphenylamine Method

Apparatus and Materials Required

- Spectrophotometer
- Cuvettes
- Pipettes
- Test tubes
- Vortex mixer
- Distilled water
- Standard DNA (e.g., calf thymus DNA)
- Saline citrate buffer (SSC)
- Glacial acetic acid
- Sulfuric acid (H_2SO_4) concentrated
- Diphenylamine
- Acetaldehyde (freshly prepared 1.6% solution)

Procedure

1. Preparation of Standard Curve:

- **Standard DNA Solution:** Prepare a stock solution of standard DNA at a known concentration (1 mg/mL) in SSC buffer.
- **Standard Dilutions:** Using serial dilutions with SSC buffer, prepare a series of standard DNA solutions with increasing quantity. A common range is 0 μg (blank), 50 μg , 100 μg , 150 μg , 200 μg , and 250 μg .

2. Diphenylamine Reagent:

- **Solution A:** Dissolve 1.5 g of diphenylamine in 100 mL of glacial acetic acid.
- **Solution B:** Carefully add 1.5 mL of concentrated sulfuric acid (H_2SO_4) to Solution A. **Warning:** Sulfuric acid is corrosive. Handle with appropriate precautions.
- **Fresh Diphenylamine Reagent:** On the day of use, prepare the working reagent by adding 0.5 mL of freshly prepared 1.6% acetaldehyde solution to 100 mL of Solution B (prepared in step 2).

3. Reaction and Measurement:

- Add 4 mL of fresh diphenylamine reagent to each test tube and mix by vortexing.
- Incubate all tubes at 60°C for 15 minutes.
- After incubation, cool the tubes down to room temperature.
- Set the spectrophotometer to a wavelength of 600 nm.
- Using a cuvette filled with distilled water as the blank, measure the absorbance of each standard solution at 600 nm.

4. Plotting the Standard Curve

- Using a graph paper, plot a standard curve with DNA (μg) on the x-axis and absorbance (650 nm) on the y-axis. This standard curve will be used to determine the protein concentration of unknown samples using their absorbance values in subsequent experiments.

5. Estimation of DNA in Unknown Sample:

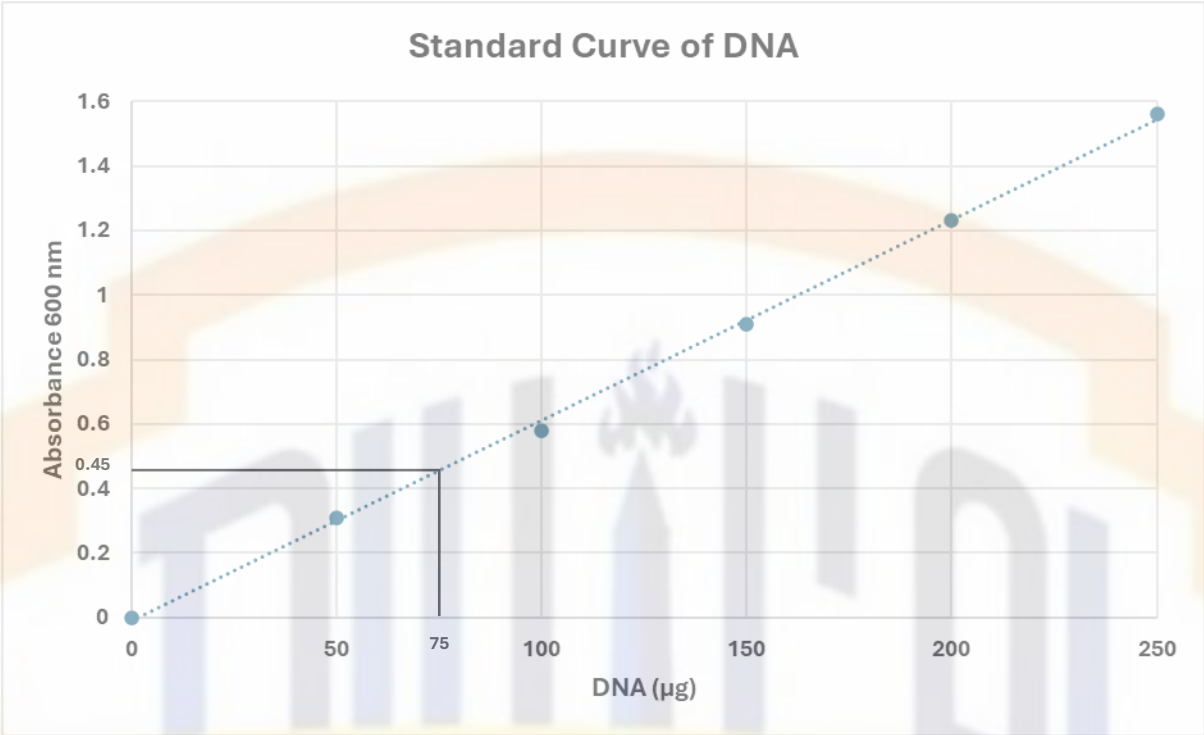
- Follow the above mentioned steps for your unknown DNA sample.
- Measure the absorbance of the unknown sample at 600 nm following the same protocol used for the standards.
- Using the standard curve (absorbance vs. DNA concentration), determine the concentration of DNA in your unknown sample.

Results

- The concentration of unknown sample is 0.45 which corresponds to 75 μg of DNA.

Sr. No.	Stock Solution (μL)	H ₂ O (μL)	Diphenyl reagent (mL)	DNA (μg)	Absorbance (600 nm)
1	0	1000	4	0	0
2	50	950	4	50	0.31
3	100	900	4	100	0.58
4	150	850	4	150	0.91
5	200	800	4	200	1.23
6	250	750	4	250	1.56

Sr. No.	Unknown Solution (μL)	H ₂ O (μL)	Diphenyl reagent (mL)	DNA (μg)	Absorbance (600 nm)
1	500	500	4	75	0.45



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PUACP

Preparation of Standard Curve and Estimation of Total RNA by Colorimetric Analysis using Orcinol Method

Apparatus and Materials Required:

- Spectrophotometer with cuvette holder
- Micropipettes (various volumes)
- Test tubes and rack
- Boiling water bath
- Timer
- Disposable gloves
- Safety glasses

Reagents:

- Standard RNA solution (known concentration)
- 1X Saline Sodium Citrate (SSC) buffer
- Orcinol solution (6% in ethanol)
- Ferric chloride solution (FeCl_3 , in concentrated HCl)
- Orcinol reagent (freshly prepared): Mix FeCl_3 solution with Orcinol solution in specific ratio
- Distilled water

Procedure:

1. Standard Curve Preparation:

- Prepare a stock solution of RNA (1mg/ mL).
- Prepare a series of test tubes labeled for standards and blank.
- Pipette increasing volumes of stock RNA solution (0, 20, 40, 60, 80 μL and 100 μL) into designated tubes.
- Adjust the volume in each tube to a constant value (200 μL) with 1X SSC buffer.
- Include a blank tube containing only 1X SSC buffer (200 μL).

2. Reaction:

- Add an equal volume (200 μL) of Orcinol reagent to all tubes (standards, sample, and blank).
- Mix the contents of each tube thoroughly by pipetting or vortexing.
- Incubate all tubes in a boiling water bath for 10-20 minutes.
- Allow the tubes to cool to room temperature.

3. Measurement:

- Set the spectrophotometer to 660 nm wavelength.
- Zero the instrument using the blank solution.
- Measure the absorbance of each standard and sample solution at 660 nm.
- Record the absorbance values for all tubes.

4. Standard Curve:

- Plot a standard curve using a graph paper.
- The X-axis represents the quantity of RNA standards ($\mu\text{g}/\text{sample volume}$).
- The Y-axis represents the absorbance values measured at 660 nm for each standard.
- The standard curve should show a linear relationship between RNA concentration and absorbance.

5. RNA Concentration in Sample:

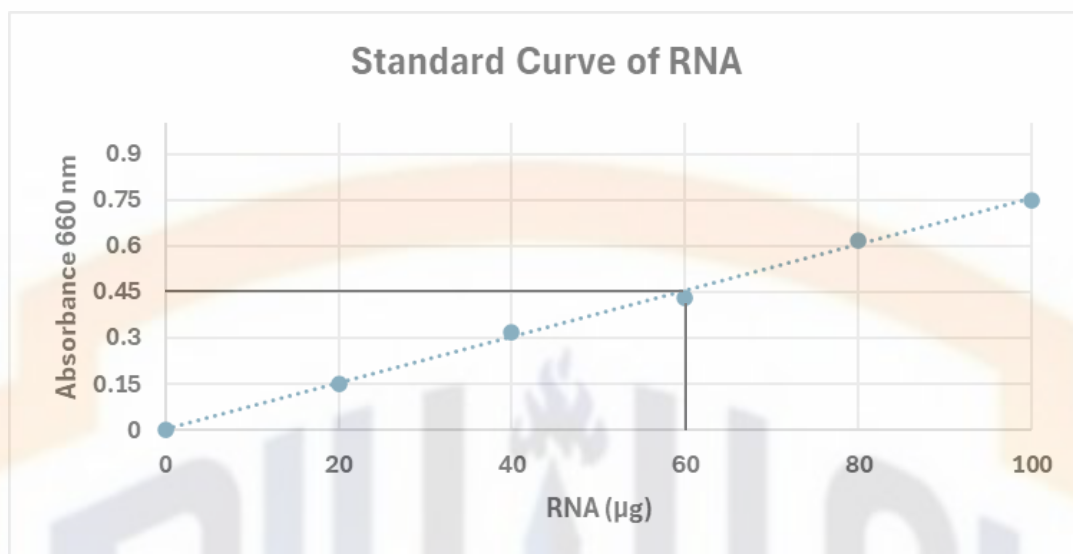
- Using the absorbance value of your sample solution, extrapolate the RNA concentration from the standard curve equation or graph.
- The concentration will be expressed in the same units used for the standards ($\mu\text{g}/\text{sample volume}$).

Results:

The absorbance of unknown sample was 45 at 660 nm which corresponds to 60 $\mu\text{g}/50 \mu\text{L}$ of RNA sample.

Sr. No.	Stock Solution (μL)	SSC buffer (μL)	Orcinol reagent (μL)	RNA (μg)	Absorbance (660 nm)
1	0	200	200	0	0
2	20	180	200	20	0.15
3	40	160	200	40	0.32
4	60	140	200	60	0.43
5	80	120	200	80	0.62
6	100	100	200	100	0.75

Sr. No.	Unknown Sample (μL)	SSC buffer (μL)	Orcinol reagent (μL)	RNA (μg)	Absorbance (660 nm)
1	50	150	200	60	45



Estimation of Free Amino Acids in Biological Samples Colorimetrically

Apparatus and Materials Required:

- Spectrophotometer with cuvette holder
- Micropipettes (various volumes)
- Test tubes and rack
- Boiling water bath
- Timer
- Disposable gloves
- Safety glasses

Reagents:

- 1 mg/ mL stock solution of a known amino acid (e.g., L-alanine) for standard curve preparation
- Buffer solution (pH 8.0 for Ninhydrin reaction)
- Ninhydrin solution
- Distilled water

Procedure:

1. Standard Curve Preparation:

- Prepare a series of test tubes labeled for standards and blank.
- Dilute the stock amino acid solution to create different quantity (0, 20, 40, 60, 80 and 100 μg).
- Adjust the volume in each tube to a constant value (200 μL) with buffer solution.
- Include a blank tube containing only buffer solution (200 μL).

2. Sample Preparation:

- Adjust the sample volume to a defined value (200 μL) with buffer solution in a separate test tube.

3. Reaction:

- Add an equal volume (200 μL) of the Ninhydrin solution to all tubes (standards, sample, and blank).
- Mix the contents of each tube thoroughly by pipetting or vortexing.
- Incubate all tubes in a boiling water bath for 20 minutes.
- Allow the tubes to cool at room temperature for 15 minutes.

4. Measurement:

- Set the spectrophotometer to 570 nm wavelength for Ninhydrin reaction.
- Zero the instrument using the blank solution.
- Measure the absorbance of each standard and sample solution at 570 nm.
- Record the absorbance values for all tubes.

Results:

1. Standard Curve:

- Plot a standard curve using a graph paper.
- The X-axis represents the concentration of amino acid standards.
- The Y-axis represents the absorbance values measured at 570 nm for each standard.
- The standard curve should show a linear relationship between amino acid concentration and absorbance.

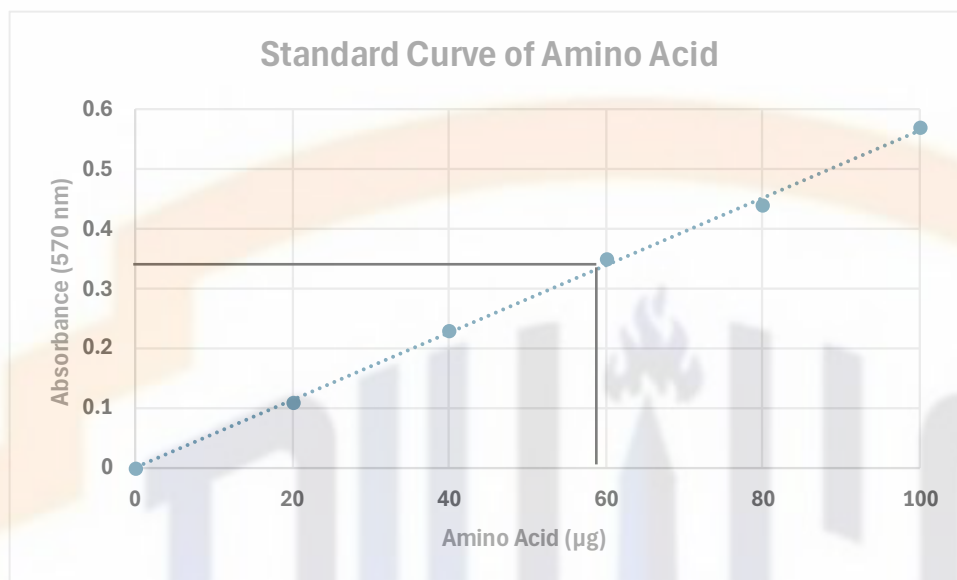
2. Free Amino Acid Concentration in Sample:

- Using the absorbance value of your sample solution, extrapolate the free amino acid concentration from the standard curve equation or graph.
- The concentration will be expressed in the same units used for the standards.

Result: The unknown sample contains 59 μg amino acids as interpolated from standard curve.

Sr. No.	Stock Solution (μL)	Buffer solution (μL)	Ninhydrin reagent (μL)	Amino Acid (μg)	Absorbance (570 nm)
1	0	200	200	0	0
2	20	180	200	20	0.11
3	40	160	200	40	0.23
4	60	140	200	60	0.35
5	80	129	200	80	0.44
6	100	100	200	100	0.57

Sr. No.	Sample (μL)	Buffer solution (μL)	Ninhydrin reagent (μL)	Amino Acid (μg)	Absorbance (570 nm)
1	50	150	200	59	0.33



Separation and Identification of Various Amino Acids by Paper Chromatography

Apparatus and Materials Required:

- Chromatography chamber (sealed tank)
- Chromatography paper (e.g., Whatman filter paper)
- Pencil
- Ruler
- Micropipettes
- Capillary tubes or microsyringes
- Spray bottle
- Developing solvent (n-butanol: acetic acid: water)
- Amino acid standards solution (mixture of known amino acids)
- Individual amino acid solutions (optional, for confirmation)
- Ninhydrin solution

Procedure:

1. Sample Preparation:

- Prepare a dilute solution of your amino acid sample (or a mixture) in distilled water.
- Prepare individual solutions of standard amino acids if confirming the identity of unknown amino acids.

2. Chromatogram Preparation:

- Cut a rectangular piece of chromatography paper.
- Using a pencil, lightly mark a line 1-2 cm from the bottom edge of the paper (start line).
- Make small pencil marks (spots) at intervals of 2 cm along the start line for sample application.

3. Sample Application:

- Dip a clean capillary tube or microsyringe into your sample solution.
- Touch the tip of the capillary tube to a designated spot on the start line, depositing a small spot of the sample solution.
- Repeat for each sample or standard solution on separate spots.
- Allow the spots to dry completely, you can use a hairdryer on a cool setting to expedite drying (optional).

4. Chromatography Development:

- Prepare the developing solvent in the chromatography chamber (ensure the chamber is saturated with solvent vapor).
- Place the chromatography paper with the spotted side facing up into the chamber.
- Ensure the solvent level in the chamber does not touch the pencil marks on the paper.

- Close the chamber securely and allow the solvent to migrate up the paper by capillary action (solvent front).
- Monitor the solvent movement. The run is complete when the solvent front reaches a designated height (e.g., 3/4th the length of the paper).
- Remove the paper from the chamber and mark the final solvent front with a pencil.

5. Detection:

- Hang the paper vertically and allow the solvent to evaporate completely.
- Prepare the Ninhydrin solution.
- Place the paper in a large tray and carefully spray the entire paper with Ninhydrin solution until it is evenly wet.
- Heat the sprayed paper using a heating plate or oven at 100°C for 5 minutes to develop colored spots corresponding to the amino acids.

6. Visualization and Identification:

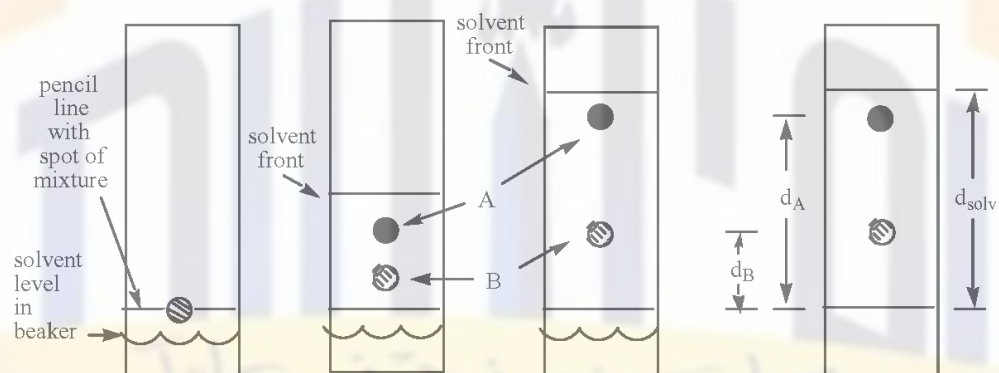
- Allow the paper to cool completely.
- Colored spots will appear on the paper, indicating the presence of amino acids.
- The distance traveled by each amino acid spot (relative to the solvent front) is characterized by its R_f value (retardation factor).
 - $R_f = \text{Distance traveled by the amino acid spot} / \text{Distance traveled by the solvent front}$
- Compare the R_f values of unknown amino acids with those of standard amino acids run on the same paper or previously established R_f values for specific amino acids and solvent systems.

Result

As the R_f value of unknown amino acid was calculated to be 0.59 which corresponds to amino acid glycine.

Amino Acid	Distance travelled by amino acid (mm)	Distance travelled by solvent front (mm)	R_f Value
Aspartic acid	31	74	0.43
Glycine	43	74	0.59
Leucine	48	74	0.65

Amino Acid	Distance travelled by amino acid (mm)	Distance travelled by solvent front (mm)	R _f Value
Unknown	43	74	0.59



Separation of Proteins by Polyacrylamide Gel Electrophoresis (PAGE)

Apparatus and Materials Required:

- Electrophoresis apparatus with buffer tank, power supply, and gel tray
- Casting stand and assembly supplies (gaskets, combs)
- Polyacrylamide gel solution (acrylamide and bis-acrylamide, appropriate concentration for desired protein size range)
- Electrode buffer (Tris-glycine or Tris-acetate-EDTA)
- Sample buffer (Laemmli buffer with reducing agent for SDS-PAGE)
- Protein standards (mixture of proteins with known molecular weights)
- Protein sample
- Staining solution (Coomassie Brilliant Blue)
- Destaining solution
- Power source

Additional requirements for specific variations of PAGE:

- Sodium dodecyl sulfate (SDS) for SDS-PAGE (to denature proteins and achieve separation based solely on size)
- Reducing agent (β -mercaptoethanol) for reducing disulfide bonds (relevant for SDS-PAGE)

Procedure:

1. Gel Preparation:

- Prepare the polyacrylamide gel solution according to the desired protein size range and total acrylamide concentration.
- Assemble the casting stand with the gel tray and a well-forming comb.
- Pour the acrylamide solution into the gel tray, leaving space for the well to form at the top.
- Overlay the solution with a layer of water or isopropanol to prevent evaporation.
- Allow the gel to polymerize.
- Remove the water/isopropanol layer and rinse the wells with electrode buffer.

2. Sample Preparation:

- Prepare your protein sample in the appropriate sample buffer. Consider protein concentration and potential requirements for denaturation and reduction (depending on the PAGE variation).
- Prepare protein standards according to the manufacturer's instructions.

3. Electrophoresis:

- Fill the buffer tank with the chosen electrode buffer.
- Carefully place the gel cassette into the electrophoresis apparatus, ensuring proper electrical orientation.

- Load your protein samples and protein standards into the wells of the gel using micropipettes.
- Run the electrophoresis at a constant voltage or current according to the gel thickness and protein size.

4. Protein Visualization:

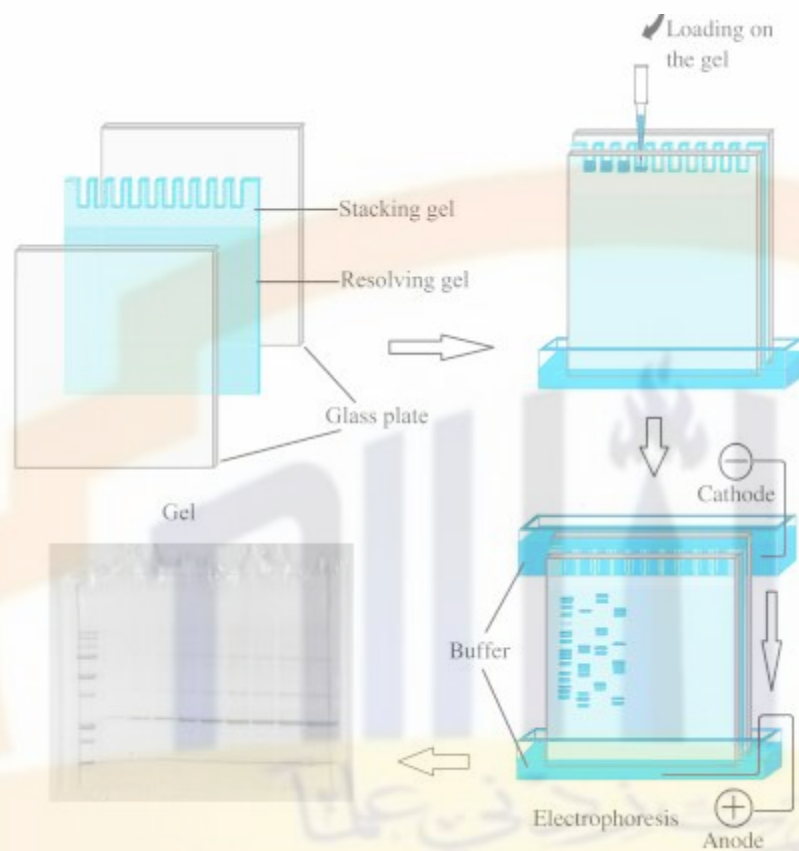
- After electrophoresis, carefully remove the gel from the cassette.
- Stain the gel with a suitable protein stain (Coomassie Brilliant Blue) following the recommended protocol.
- The staining process allows visualization of protein bands corresponding to the separated proteins in the sample and standards.
- Destaining might be required for further analysis.

Analysis:

- The protein separation will be visualized as distinct bands on the stained gel.
- Larger proteins migrate slower through the gel and appear lower on the gel, while smaller proteins migrate faster and appear higher on the gel (applies to standard PAGE).
- By comparing the migration pattern of your protein sample to the protein standards with known molecular weights, you can estimate the size of the proteins in your sample.

Result:

The proteins were separated from a mixture by SDS-PAGE. Three proteins at 65, 70 and 110 kDa were observed on the gel.



PUACP

Quantitative Analysis of Amylase Activity from Blood Serum or Liver

Materials and Apparatus Required:

Blood Serum Sample: Obtain a fresh blood serum sample.

Centrifuge: To separate serum from blood cells.

Test Tubes or Microplates: For performing the enzymatic assay.

Amylase Substrate Solution: A starch solution (e.g., 1% soluble starch).

Iodine Solution: To detect the presence of starch.

pH Buffer Solution: To maintain a specific pH for the enzymatic reaction.

Amylase Standard Solution: A known concentration of amylase for calibration.

Spectrophotometer: To measure absorbance at a specific wavelength.

Cuvettes: For holding the reaction mixture during spectrophotometric analysis.

Stopwatch or Timer: To monitor reaction time.

Procedure:

1. Sample Preparation:

- Collect a blood sample and allow it to clot.
- Centrifuge the blood to separate serum from blood cells.

2. Enzymatic Assay:

- Prepare a series of test tubes or microplates.
- In each tube, mix the following:
 - Serum sample
 - Amylase substrate solution (Starch)
 - pH buffer solution
- Incubate the tubes at 37°C for 10 minutes.

3. Stopping the Reaction:

- Add iodine solution to each tube to stop the enzymatic reaction.
- The iodine reacts with any remaining starch, forming a blue-black color.

4. Spectrophotometric Analysis:

- Measure the absorbance of each tube at a specific wavelength (540 nm).
- Use a blank (without amylase) to calibrate the spectrophotometer.

5. Calculations:

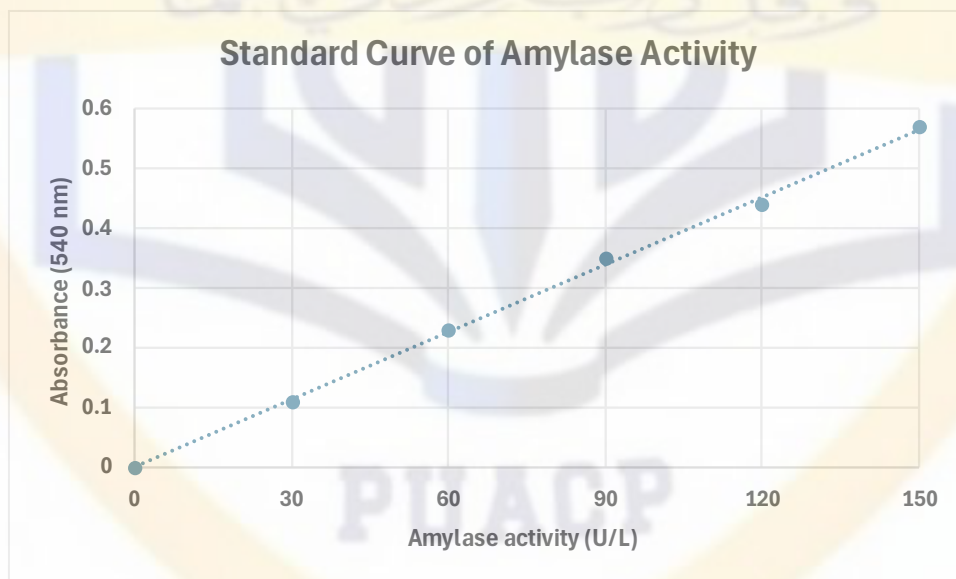
- Construct a standard curve using the amylase standard solution (0, 30, 60, 90, 120 and 150 U/L).
- Determine the amylase activity in the sample based on the absorbance and the standard curve.

Results:

The absorbance of sample was 0.23 which corresponds to 60 U/L of amylase activity according to the standard curve. This activity is within the normal range i.e., 25-125 U/L.

Sr. No.	Amylase (U/ L)	Absorbance (540 nm)
1	0	0
2	30	0.11
3	60	0.23
4	90	0.35
5	120	0.44
6	150	0.57

Serum Amylase (U/ L)	Absorbance (540 nm)
60	0.23



Effect of pH on the Enzymatic Activity of Salivary Amylase

Apparatus and Materials Required:

Starch

Amylase from Saliva

Test Tubes

NaOH

Iodine solution

Fehling's Solution

Spectrophotometer

Buffer Solution

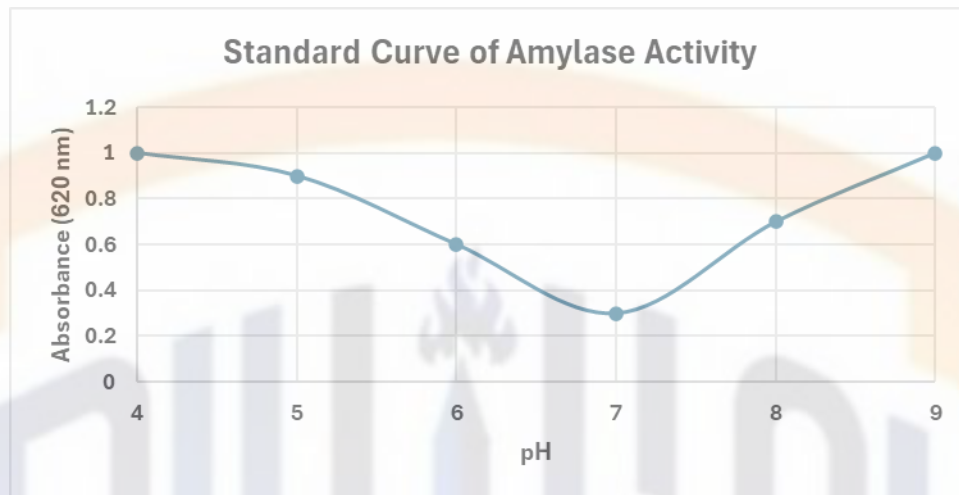
Procedure

1. Pour 5 mL of buffer (pH 4, 6, 9) in different test tubes.
2. Add 5 mL of starch solution (20 g/L) and 1 mL of saliva to each tube except the control.
3. Incubate test tubes at 25 °C for 10 minutes in a water bath.
4. Take 0.5 mL of the reaction mixture and add 5 mL of 0.1 N HCl to stop the reaction.
5. Add a few drops of iodine solution to the stopped reaction.
6. Blank will show bluish-black color while brown-red will be shown for partially degraded starch, and no color for totally degraded starch.
7. Take absorbance at 620 nm to check the optimum pH for enzyme activity.

Result

The lowest absorbance was found at pH 7 which corresponds to the lowest concentration of starch or highest activity of amylase.

Sr. No.	pH	Absorbance (620 nm)
1	4	1
2	5	0.9
3	6	0.6
4	7	0.3
5	8	0.7
6	9	1



Effect of Temperature on the Enzymatic Activity of Salivary Amylase

Materials and Apparatus Required:

Starch

Amylase from Saliva

Test Tubes

NaOH

Iodine solution

Fehling's Solution

Spectrophotometer

Buffer Solution

Procedure:

1. Adjust water temperature in 1 L beaker (30 to 90 °C).
2. Pour 5 mL of buffer (pH 7.0) in 5 mL of starch solution (20 g/ L).
3. Aliquot 2 mL in five test tubes and incubate test tubes at different temperatures for 5 minutes.
4. Add 0.2 mL of human saliva solution in each tube.
5. Incubate for another 10 minutes.
6. Add 1 mL of 0.1 N HCl to stop the reaction.
7. Dispense a few drops of Iodine solution to the stopped reaction.
8. It will be turned into blue for unreacted starch molecules, brownish red for partially degraded starch, and no color for a completely hydrolyzed starch.
9. Note the absorbance at 620 nm.

Result:

The lowest absorbance was found at 37 °C which corresponds to the lowest concentration of starch or highest activity of amylase.

Sr. No.	Temperature (°C)	Absorbance (620 nm)
1	20	1
2	25	0.9
3	30	0.8
4	35	0.3
5	40	1

